

```
<!DOCTYPE html>
<html lang="ar" dir="rtl">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>MESH SPX Command Center | مركز قيادة SPX</title>
  <style>
    * { margin: 0; padding: 0; box-sizing: border-box; }

    body {
      background: #050a0f;
      color: #e0e6ed;
      font-family: 'Segoe UI', Tahoma, sans-serif;
      min-height: 100vh;
    }

    .header {
      background: linear-gradient(135deg, #0a1628 0%, #0d1f3c 100%);
      border-bottom: 1px solid #1a3a5c;
      padding: 14px 20px;
      display: flex;
      justify-content: space-between;
      align-items: center;
    }
    .header-title h1 { font-size: 20px; color: #00d4ff; letter-spacing: 1px; }
    .header-title p { font-size: 12px; color: #5a7a9a; margin-top: 2px; }
    .header-right { text-align: left; font-size: 11px; color: #5a7a9a; }
    .refresh-btn {
      background: #0a2040; border: 1px solid #1a4a7a; color: #00d4ff;
      padding: 7px 14px; border-radius: 8px; cursor: pointer; font-size: 12px;
      margin-top: 5px; display: block;
    }

    .judas-bar {
      padding: 9px 20px; font-size: 12px;
      display: flex; justify-content: space-between; align-items: center;
      border-bottom: 1px solid #1a3a5c;
    }
    .judas-bar.active { background: rgba(255,170,0,0.1); border-color: #ffaa00;
color: #ffaa00; }
    .judas-bar.inactive { background: rgba(0,212,255,0.05); color: #5a7a9a; }

    .main-grid {
      display: grid;

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    grid-template-columns: 1fr 1fr;
    gap: 14px;
    padding: 14px 20px;
}
@media (max-width: 700px) { .main-grid { grid-template-columns: 1fr; } }

.card {
    background: #0a1628;
    border: 1px solid #1a3a5c;
    border-radius: 12px;
    padding: 14px;
}
.card-title {
    font-size: 10px; color: #5a7a9a;
    text-transform: uppercase; letter-spacing: 1px;
    margin-bottom: 12px;
    display: flex; justify-content: space-between;
}

.price-box {
    background: #050f1f; border-radius: 10px;
    padding: 14px; text-align: center; margin-bottom: 12px;
}
.price-label { font-size: 10px; color: #5a7a9a; margin-bottom: 4px; }
.price-value { font-size: 34px; color: #00d4ff; font-weight: 700; letter-
spacing: 1px; }

.gex-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 8px; margin-
top: 8px; }
.gex-item { background: #050f1f; border-radius: 8px; padding: 10px 12px; }
.gex-item-label { font-size: 10px; color: #5a7a9a; margin-bottom: 4px; }
.gex-item-value { font-size: 19px; font-weight: 700; }
.call-wall { color: #00c853; border-right: 3px solid #00c853; padding-right:
8px; }
.put-wall { color: #ff1744; border-right: 3px solid #ff1744; padding-right:
8px; }
.zero-gamma { color: #ff9800; border-right: 3px solid #ff9800; padding-right:
8px; }
.max-pain { color: #29b6f6; border-right: 3px solid #29b6f6; padding-right:
8px; }
.gamma-flip { color: #e040fb; }

.move-box { border-radius: 10px; padding: 12px; margin-bottom: 8px; }
.move-daily { background: rgba(0,212,255,0.08); border: 1px solid
rgba(0,212,255,0.2); }
.move-weekly { background: rgba(224,64,251,0.08); border: 1px solid
rgba(224,64,251,0.2); }

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    .move-title { font-size: 10px; margin-bottom: 8px; display: flex; justify-
content: space-between; }
    .move-daily .move-title { color: #00d4ff; }
    .move-weekly .move-title { color: #e040fb; }
    .move-values { display: grid; grid-template-columns: 1fr 1fr; gap: 8px; }
    .move-up, .move-down { background: #050f1f; border-radius: 8px; padding: 8px;
text-align: center; }
    .move-up { border-top: 2px solid #00c853; }
    .move-down { border-top: 2px solid #ff1744; }
    .move-dir { font-size: 10px; color: #5a7a9a; }
    .move-val { font-size: 17px; font-weight: 700; color: #e0e6ed; }

    .vix-big { font-size: 40px; font-weight: 700; color: #e0e6ed; }
    .vix-change { font-size: 12px; padding: 3px 10px; border-radius: 6px; display:
inline-block; margin-top: 4px; }
    .vix-label { font-size: 10px; color: #5a7a9a; margin-top: 4px; }

    .iv-row { margin-top: 12px; }
    .iv-label-row { display: flex; justify-content: space-between; font-size:
11px; margin-bottom: 4px; }
    .iv-bar-bg { background: #050f1f; border-radius: 4px; height: 6px; overflow:
hidden; }
    .iv-bar-fill { height: 100%; border-radius: 4px; background: linear-
gradient(90deg, #00d4ff, #e040fb); transition: width 0.5s; }

    .strategy-box {
    margin-top: 12px; background: rgba(224,64,251,0.1);
    border: 1px solid rgba(224,64,251,0.3); border-radius: 10px;
    padding: 12px; text-align: center;
    }
    .strategy-label { font-size: 10px; color: #5a7a9a; margin-bottom: 4px; }
    .strategy-value { font-size: 15px; color: #e040fb; font-weight: 700; }
    .strategy-ar { font-size: 12px; color: #a060c0; margin-top: 2px; }

    .pc-box {
    background: #050f1f; border-radius: 10px; padding: 12px;
    display: flex; justify-content: space-between; align-items: center; margin-
bottom: 8px;
    }
    .pc-label { font-size: 10px; color: #5a7a9a; }
    .pc-value { font-size: 22px; font-weight: 700; }
    .pc-signal { font-size: 10px; margin-top: 2px; }

    .gex-regime-box { margin-top: 10px; padding: 10px; border-radius: 8px; }

    .bias-box { text-align: center; padding: 14px; border-radius: 10px; margin-
bottom: 12px; }

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```
.bias-neutral { background: rgba(90,120,150,0.1); border: 1px solid
rgba(90,120,150,0.3); }
.bias-bullish { background: rgba(0,200,83,0.1); border: 1px solid
rgba(0,200,83,0.3); }
.bias-bearish { background: rgba(255,23,68,0.1); border: 1px solid
rgba(255,23,68,0.3); }
.bias-value { font-size: 26px; font-weight: 700; letter-spacing: 2px; }
.bias-ar { font-size: 15px; margin-top: 4px; }
.bias-neutral .bias-value, .bias-neutral .bias-ar { color: #90a0b0; }
.bias-bullish .bias-value, .bias-bullish .bias-ar { color: #00c853; }
.bias-bearish .bias-value, .bias-bearish .bias-ar { color: #ff1744; }
.bias-strength { font-size: 11px; color: #5a7a9a; margin-top: 6px; }
.bias-reason { font-size: 11px; color: #7a9ab0; margin-top: 8px; line-height:
1.5; }
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```
.trade-today { padding: 10px 14px; border-radius: 10px; margin-bottom: 12px;
font-size: 13px; font-weight: 700; text-align: center; }
.trade-yes { background: rgba(0,200,83,0.1); border: 1px solid
rgba(0,200,83,0.3); color: #00c853; }
.trade-no { background: rgba(255,23,68,0.1); border: 1px solid
rgba(255,23,68,0.3); color: #ff1744; }
```

```
.decision-box { background: rgba(0,212,255,0.05); border: 1px solid
rgba(0,212,255,0.2); border-radius: 12px; padding: 14px; margin-bottom: 14px; }
.decision-label { font-size: 10px; color: #5a7a9a; margin-bottom: 6px; }
.decision-value { font-size: 18px; font-weight: 700; color: #00d4ff; }
.decision-detail { font-size: 12px; color: #7a9ab0; margin-top: 6px; }
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```
.liq-grid { display: grid; grid-template-columns: 1fr 1fr; gap: 8px; margin-
bottom: 10px; }
.liq-item { background: #050f1f; border-radius: 8px; padding: 10px; text-
align: center; }
.liq-label { font-size: 10px; color: #5a7a9a; margin-bottom: 4px; }
.liq-value { font-size: 16px; font-weight: 700; color: #e0e6ed; }
```

```
.scenario-item { background: #050f1f; border-radius: 10px; padding: 12px;
margin-bottom: 8px; }
.scenario-header { display: flex; justify-content: space-between; align-items:
center; margin-bottom: 8px; }
.scenario-name { font-size: 12px; font-weight: 600; }
.scenario-prob { font-size: 15px; font-weight: 700; }
.scenario-bar-bg { background: #0a1628; border-radius: 4px; height: 4px;
margin-bottom: 8px; overflow: hidden; }
.scenario-bar-fill { height: 100%; border-radius: 4px; }
.scenario-details { display: grid; grid-template-columns: 1fr 1fr; gap: 8px;
font-size: 11px; }
.scenario-target { color: #00c853; }
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.scenario-stop { color: #ff1744; }
.bullish .scenario-name, .bullish .scenario-prob { color: #00c853; }
.bullish .scenario-bar-fill { background: #00c853; }
.neutral-s .scenario-name, .neutral-s .scenario-prob { color: #90a0b0; }
.neutral-s .scenario-bar-fill { background: #90a0b0; }
.bearish .scenario-name, .bearish .scenario-prob { color: #ff1744; }
.bearish .scenario-bar-fill { background: #ff1744; }

.cal-table { width: 100%; border-collapse: collapse; font-size: 11px; }
.cal-table th { color: #5a7a9a; padding: 6px 8px; text-align: right; border-
bottom: 1px solid #1a3a5c; font-size: 10px; }
.cal-table td { padding: 8px; border-bottom: 1px solid #0a2040; }
.impact-high { background: rgba(255,23,68,0.15); color: #ff1744; padding: 2px
8px; border-radius: 4px; font-size: 10px; font-weight: 700; }
.impact-medium { background: rgba(255,152,0,0.15); color: #ff9800; padding:
2px 8px; border-radius: 4px; font-size: 10px; }

.key-level-row { display: flex; justify-content: space-between; align-items:
center; padding: 8px 0; border-bottom: 1px solid #0a2040; font-size: 12px; }
.level-price { font-weight: 700; min-width: 60px; }
.level-name { color: #7a9ab0; font-size: 11px; flex: 1; margin: 0 10px; }
.level-strength { font-size: 10px; padding: 2px 8px; border-radius: 4px; }
.strong { background: rgba(255,170,0,0.15); color: #ffaa00; }
.moderate { background: rgba(90,120,150,0.15); color: #90a0b0; }

.loading { display: flex; align-items: center; justify-content: center;
padding: 20px; color: #5a7a9a; font-size: 12px; gap: 8px; }
.spinner { width: 16px; height: 16px; border: 2px solid #1a3a5c; border-top-
color: #00d4ff; border-radius: 50%; animation: spin 0.8s linear infinite; }
@keyframes spin { to { transform: rotate(360deg); } }

.full-width { grid-column: 1 / -1; }
.footer { text-align: center; padding: 14px; color: #2a4a6a; font-size: 10px;
border-top: 1px solid #1a3a5c; margin-top: 8px; }

.market-closed { background: rgba(255,152,0,0.1); border: 1px solid
rgba(255,152,0,0.3); color: #ff9800; padding: 6px 14px; border-radius: 8px; font-
size: 11px; display: inline-block; margin-top: 4px; }
</style>
</head>
<body>

<div class="header">
  <div class="header-title">
    <h1>⚡ MESH SPX Command Center</h1>
    <p>نظام التداول اليومي - SPX مركز قيادة</p>
  </div>

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<div class="header-right">
  <div id="last-update">جاري التحميل...</div>
  <div id="next-refresh"></div>
  <div id="market-status"></div>
  <button class="refresh-btn" onclick="loadAllData()"><img alt="refresh icon" data-bbox="638 121 658 138"/> تحديث</button>
</div>
</div>

<div class="judas-bar inactive" id="judas-bar">
  <span id="judas-text"><img alt="judas icon" data-bbox="338 218 358 235"/> لا يوجد انحراف خادع (جوداس)</span>
  <span id="judas-text-en">No Judas Swing detected</span>
</div>

<div class="main-grid">

  <!-- SPX KEY LEVELS -->
  <div class="card">
    <div class="card-title">
      <span>SPX Key Levels</span>
      <span>المستويات الرئيسية</span>
    </div>
    <div class="price-box">
      <div class="price-label">CURRENT PRICE / السعر الحالي</div>
      <div class="price-value" id="spx-price"><div class="loading"><div
class="spinner"></div></div></div>
    </div>
    <div class="gex-grid">
      <div class="gex-item">
        <div class="gex-item-label">MAX PAIN - الحد الأقصى للألم</div>
        <div class="gex-item-value max-pain" id="max-pain"></div>
      </div>
      <div class="gex-item">
        <div class="gex-item-label">ZERO GAMMA - جاما الصفرية</div>
        <div class="gex-item-value zero-gamma" id="zero-gamma"></div>
      </div>
      <div class="gex-item">
        <div class="gex-item-label">CALL WALL - حائط الشراء</div>
        <div class="gex-item-value call-wall" id="call-wall"></div>
      </div>
      <div class="gex-item">
        <div class="gex-item-label">PUT WALL - حائط البيع</div>
        <div class="gex-item-value put-wall" id="put-wall"></div>
      </div>
    </div>
    <div class="gex-item" style="margin-top:8px;">
      <div class="gex-item-label">GAMMA FLIP - نقطة التحول</div>
      <div class="gex-item-value gamma-flip" id="gamma-flip"></div>
    </div>
  </div>

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</div>
<div style="margin-top:12px;">
  <div class="move-box move-daily">
    <div class="move-title"><span>● DAILY EXPECTED MOVE</span>
<span>اليوم</span></div>
    <div class="move-values">
      <div class="move-up"><div class="move-dir">↗ UPPER</div><div
class="move-val" id="daily-upper"></div></div>
      <div class="move-down"><div class="move-dir">↘ LOWER</div><div
class="move-val" id="daily-lower"></div></div>
    </div>
  </div>
  <div class="move-box move-weekly">
    <div class="move-title"><span>● WEEKLY EXPECTED MOVE</span>
<span>الأسبوعي</span></div>
    <div class="move-values">
      <div class="move-up"><div class="move-dir">↗ UPPER</div><div
class="move-val" id="weekly-upper"></div></div>
      <div class="move-down"><div class="move-dir">↘ LOWER</div><div
class="move-val" id="weekly-lower"></div></div>
    </div>
    <div style="text-align:center;font-size:10px;color:#5a7a9a;margin-
top:6px;" id="exp-date"></div>
  </div>
</div>

<!-- VOLATILITY -->
<div class="card">
  <div class="card-title"><span>Volatility & IV</span><span>التقلب والتقلب
الضمني</span></div>
  <div class="vix-big" id="vix-value"></div>
  <span class="vix-change" id="vix-change"
style="background:rgba(90,120,150,0.2);color:#90a0b0;">>0.00 (0.00%)</span>
  <div class="vix-label">VIX LEVEL - مستوى مؤشر الخوف</div>
  <div class="iv-row">
    <div class="iv-label-row"><span>IV RANK</span><span id="iv-rank-val"
style="color:#00d4ff;font-weight:700;"></span></div>
    <div class="iv-bar-bg"><div class="iv-bar-fill" id="iv-rank-bar"
style="width:0%"></div></div>
  </div>
  <div class="iv-row" style="margin-top:10px;">
    <div class="iv-label-row"><span>IV PERCENTILE</span><span id="iv-pct-val"
style="color:#e040fb;font-weight:700;"></span></div>
    <div class="iv-bar-bg"><div class="iv-bar-fill" id="iv-pct-bar"
style="width:0%;background:linear-gradient(90deg,#e040fb,#ff1744)"></div></div>
  </div>

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<div class="pc-box" style="margin-top:12px;">
  <div>
    <div class="pc-label">PUT/CALL RATIO</div>
    <div class="pc-value" id="pc-ratio">—</div>
    <div class="pc-signal" id="pc-signal" style="color:#5a7a9a;">—</div>
  </div>
  <div style="font-size:26px;" id="pc-icon">🇮🇹</div>
</div>
<div class="strategy-box">
  <div class="strategy-label">STRATEGY RECOMMENDATION — الاستراتيجية الموصى بها</div>
  <div class="strategy-value" id="strategy-en">—</div>
  <div class="strategy-ar" id="strategy-ar">—</div>
</div>
<div class="gex-regime-box" id="gex-regime-box"
style="background:rgba(255,23,68,0.08);border:1px solid rgba(255,23,68,0.2);">
  <div style="font-size:10px;color:#5a7a9a;margin-bottom:4px;">GEX
REGIME</div>
  <div style="font-size:15px;font-weight:700;" id="gex-regime-text">—</div>
</div>
</div>

<!-- BIAS & DECISION -->
<div class="card">
  <div class="card-title"><span>Market Bias & Decision</span><span>اتجاه السوق والقرار</span></div>
  <div class="trade-today trade-yes" id="trade-today"><div class="loading"><div
class="spinner"></div></div></div>
  <div class="bias-box bias-neutral" id="bias-box">
    <div class="bias-value" id="bias-value">—</div>
    <div class="bias-ar" id="bias-ar">—</div>
    <div class="bias-strength" id="bias-strength">—</div>
    <div class="bias-reason" id="bias-reason">—</div>
  </div>
  <div class="decision-box">
    <div class="decision-label">🎯 القرار الحاسم اليوم</div>
    <div class="decision-value" id="decision-value">—</div>
    <div class="decision-detail" id="decision-detail">—</div>
  </div>
  <div class="card-title" style="margin-top:4px;"><span>Liquidity Levels</span>
<span>مستويات السيولة</span></div>
  <div class="liq-grid">
    <div class="liq-item"><div class="liq-label">PDH — أعلى</div><div
class="liq-value" id="pdh">—</div></div>
    <div class="liq-item"><div class="liq-label">PDL — أدنى</div><div
class="liq-value" id="pdl">—</div></div>
    <div class="liq-item"><div class="liq-label">WEEKLY HIGH</div><div

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class="liq-value" id="weekly-high"></div></div>
    <div class="liq-item"><div class="liq-label">WEEKLY LOW</div><div
class="liq-value" id="weekly-low"></div></div>
    </div>
</div>

<!-- SCENARIOS -->
<div class="card">
    <div class="card-title"><span>Trading Scenarios</span><span>سيناريوهات
التداول</span></div>
    <div id="scenarios-container"><div class="loading"><div class="spinner"></div>
</div></div>
</div>

<!-- CALENDAR -->
<div class="card full-width">
    <div class="card-title"><span><img alt="Saudi Arabia flag" data-bbox="431 353 451 368"/> Economic Calendar</span><span>التقويم
الاقتصادي</span></div>
    <div id="calendar-container"><div class="loading"><div class="spinner"></div>
</div></div>
</div>

<!-- KEY LEVELS -->
<div class="card full-width">
    <div class="card-title"><span>Key Support & Resistance</span><span>الدعم
والمقاومة الرئيسية</span></div>
    <div id="key-levels-container"><div class="loading"><div class="spinner">
</div></div></div>
</div>

</div>

<div class="footer">MESH SPX Command Center v2.1 | تحديث كل 15
دقيقة</div>

<script>
// =====
// CLIENT-SIDE DATA FETCHING
// المتصفح يسحب البيانات مباشرة
// =====

const CORS_PROXY = 'https://corsproxy.io/?url=';

let refreshTimer;
let nextRefreshTime;
let cachedData = null;

```

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// — FETCH SPX PRICE —
async function fetchSPX() {
  const urls = [
    'https://query1.finance.yahoo.com/v8/finance/chart/%5EGSPC?
interval=1m&range=1d',
    'https://query2.finance.yahoo.com/v8/finance/chart/%5EGSPC?
interval=1m&range=1d',
  ];
  for (const url of urls) {
    try {
      const res = await fetch(CORS_PROXY + encodeURIComponent(url));
      const data = await res.json();
      const price = data?.chart?.result?.[0]?.meta?.regularMarketPrice;
      if (price && price > 1000) return { price };
    } catch {}
  }
  // Fallback: use last known price
  return { price: cachedData?.spx?.price || 6582, isStale: true };
}

// — FETCH VIX —
async function fetchVIX() {
  const urls = [
    'https://query1.finance.yahoo.com/v8/finance/chart/%5EVIX?
interval=1d&range=1y',
    'https://query2.finance.yahoo.com/v8/finance/chart/%5EVIX?
interval=1d&range=1y',
  ];
  for (const url of urls) {
    try {
      const res = await fetch(CORS_PROXY + encodeURIComponent(url));
      const data = await res.json();
      const result = data?.chart?.result?.[0];
      const closes = result?.indicators?.quote?.[0]?.close?.filter(Boolean) || [];
      const value = result?.meta?.regularMarketPrice;
      if (!value) continue;
      const prevClose = result?.meta?.previousClose || value;
      const change = parseFloat((value - prevClose).toFixed(2));
      const changePct = parseFloat(((change / prevClose) * 100).toFixed(2));
      const low52 = closes.length > 50 ? Math.min(...closes.slice(-252)) : 12;
      const high52 = closes.length > 50 ? Math.max(...closes.slice(-252)) : 45;
      const percentile = closes.length > 10
        ? Math.round(closes.filter(c => c < value).length / closes.length * 100)
        : 60;
      return { value, change, changePct, low52, high52, percentile };
    } catch {}
  }
}

```

```
    return cachedData?.vix || { value: 23.87, change: 0, changePct: 0, low52: 12,
high52: 45, percentile: 60 };
}
```

```
// — FETCH OPTIONS DATA —
```

```
async function fetchOptions(spxPrice) {
  const url = 'https://query1.finance.yahoo.com/v7/finance/options/%5ESPX';
  try {
    const res = await fetch(CORS_PROXY + encodeURIComponent(url));
    const data = await res.json();
    const result = data?.optionChain?.result?.[0];
    if (!result) return null;

    const calls = result.options?.[0]?.calls || [];
    const puts = result.options?.[0]?.puts || [];
    if (!calls.length || !puts.length) return null;

    const maxPain = calcMaxPain(calls, puts);
    const gex = calcGEX(calls, puts, spxPrice);
    const totalCallOI = calls.reduce((s, c) => s + (c.openInterest || 0), 0);
    const totalPutOI = puts.reduce((s, p) => s + (p.openInterest || 0), 0);
    const ratio = totalCallOI > 0 ? parseFloat((totalPutOI /
totalCallOI).toFixed(2)) : 0.87;
    return { gex, maxPain, putCall: { ratio } };
  } catch {
    return null;
  }
}
```

```
// — FETCH LIQUIDITY —
```

```
async function fetchLiquidity() {
  const urls = [
    'https://query1.finance.yahoo.com/v8/finance/chart/%5EGSPC?
interval=1d&range=10d',
    'https://query2.finance.yahoo.com/v8/finance/chart/%5EGSPC?
interval=1d&range=10d',
  ];
  for (const url of urls) {
    try {
      const res = await fetch(CORS_PROXY + encodeURIComponent(url), {
        headers: { 'Accept': 'application/json' }
      });
      const data = await res.json();
      const result = data?.chart?.result?.[0];
      const quotes = result?.indicators?.quote?.[0];
      const highs = quotes?.high?.filter(Boolean) || [];
      const lows = quotes?.low?.filter(Boolean) || [];
```

```

    const closes = quotes?.close?.filter(Boolean) || [];
    if (highs.length < 2) continue;
    return {
      pdh: parseFloat(highs[highs.length - 2].toFixed(2)),
      pdl: parseFloat( lows[ lows.length - 2].toFixed(2)),
      weeklyHigh: parseFloat(Math.max(...highs.slice(-5)).toFixed(2)),
      weeklyLow:  parseFloat(Math.min(...lows.slice(-5)).toFixed(2)),
    };
  } catch {}
}

// Fallback: estimate from current price
const price = cachedData?.spx?.price || 6582;
return {
  pdh: parseFloat((price * 1.004).toFixed(2)),
  pdl: parseFloat((price * 0.996).toFixed(2)),
  weeklyHigh: parseFloat((price * 1.008).toFixed(2)),
  weeklyLow:  parseFloat((price * 0.985).toFixed(2)),
};
}

// =====
//   CALCULATIONS
// =====

function calcMaxPain(calls, puts) {
  const strikes = [...new Set([...calls.map(c => c.strike), ...puts.map(p => p.strike)])].sort((a, b) => a - b);
  let minLoss = Infinity, mpStrike = strikes[Math.floor(strikes.length / 2)];
  for (const s of strikes) {
    let loss = 0;
    for (const c of calls) if (c.strike < s) loss += (s - c.strike) * (c.openInterest || 0);
    for (const p of puts) if (p.strike > s) loss += (p.strike - s) * (p.openInterest || 0);
    if (loss < minLoss) { minLoss = loss; mpStrike = s; }
  }
  return mpStrike;
}

function calcGEX(calls, puts, price) {
  const callsAbove = calls.filter(c => c.strike > price).sort((a, b) => (b.openInterest || 0) - (a.openInterest || 0));
  const putsBelow = puts.filter(p => p.strike < price).sort((a, b) => (b.openInterest || 0) - (a.openInterest || 0));
  const callWall = callsAbove[0]?.strike || Math.round(price * 1.02 / 25) * 25;
  const putWall = putsBelow[0]?.strike || Math.round(price * 0.98 / 25) * 25;
  const zeroGamma = Math.round((callWall + putWall) / 2 / 5) * 5;

```

```

    const gammaFlip = Math.round((zeroGamma + price) / 2 / 5) * 5;
    const netGex = callsAbove.slice(0, 5).reduce((s, c) => s + (c.openInterest ||
0), 0)
        - putsBelow.slice(0, 5).reduce((s, p) => s + (p.openInterest || 0),
0);
    return { callWall, putWall, zeroGamma, gammaFlip, netGex };
}

function fallbackGEX(price) {
    return {
        callWall: Math.round(price * 1.015 / 25) * 25,
        putWall: Math.round(price * 0.985 / 25) * 25,
        zeroGamma: Math.round(price / 25) * 25,
        gammaFlip: Math.round(price * 1.005 / 5) * 5,
        netGex: -1,
    };
}

function ivRank(vix, low = 12, high = 45) {
    if (!vix || high === low) return 30;
    return Math.min(100, Math.max(0, Math.round(((vix - low) / (high - low)) *
100)));
}

function getStrategy(rank) {
    if (rank < 20) return { en: 'Do Not Trade', ar: 'لا تتداول اليوم', key:
'no_trade' };
    if (rank < 40) return { en: 'Credit Spread Narrow', ar: 'سبريد ائتماني ضيق',
key: 'cs_narrow' };
    if (rank < 60) return { en: 'Iron Condor', ar: 'آيرون كوندور', key: 'ic' };
    if (rank < 80) return { en: 'Iron Condor Wide', ar: 'آيرون كوندور واسع', key:
'ic_wide' };
    return { en: 'Iron Fly', ar: 'آيرون فلاي', key: 'fly' };
}

function getBias(price, gex, vix) {
    const aboveZero = price > gex.zeroGamma;
    const negGex = gex.netGex < 0;
    if (aboveZero && !negGex) return { direction: 'bullish', strength: 65, reason:
`إيجابي - استقرار مفضل GEX Zero-Gamma (${gex.zeroGamma}). السعر فوق` };
    if (!aboveZero && negGex) return { direction: 'bearish', strength: 68, reason:
`سلبي - يضغط الحركة الهابطة GEX Zero-Gamma (${gex.zeroGamma}). السعر تحت` };
    return { direction: 'neutral', strength: vix > 25 ? 42 : 52, reason: `السعر بين
Put Wall (${gex.putWall}) و Call Wall (${gex.callWall}). Max Pain مغناطيس
${gex.maxPain || Math.round(price/25)*25}.` };
}

```

```

function getExpectedMove(price, vix) {
    const d = Math.round(price * (vix / 100) * Math.sqrt(1 / 365));
    const w = Math.round(price * (vix / 100) * Math.sqrt(5 / 365));
    const today = new Date();
    const fri = new Date(today);
    fri.setDate(today.getDate() + ((5 - today.getDay() + 7) % 7 || 7));
    return { dailyUpper: price + d, dailyLower: price - d, weeklyUpper: price + w,
weeklyLower: price - w, expDate: fri.toISOString().split('T')[0] };
}

function getScenarios(price, gex, dir) {
    const cw = gex.callWall, pw = gex.putWall, mp = gex.maxPain ||
Math.round(price/25)*25, zg = gex.zeroGamma;
    if (dir === 'bearish') return [
        { type: 'bearish', icon: '↘', name: 'Bearish Continuation', nameAr: 'هبوط مستمر', prob: 55, target: pw, stop: zg },
        { type: 'neutral-s', icon: '↔', name: 'Range-Bound', nameAr: 'تداول في نطاق', prob: 30, target: mp, stop: Math.round(pw * 0.995) },
        { type: 'bullish', icon: '↗', name: 'Bullish Reversal', nameAr: 'انعكاس صاعد', prob: 15, target: zg, stop: pw },
    ];
    if (dir === 'bullish') return [
        { type: 'bullish', icon: '↗', name: 'Bullish Breakout', nameAr: 'اختراق صعودي', prob: 50, target: cw, stop: zg },
        { type: 'neutral-s', icon: '↔', name: 'Range-Bound', nameAr: 'تداول في نطاق', prob: 35, target: mp, stop: pw },
        { type: 'bearish', icon: '↘', name: 'Bearish Flush', nameAr: 'هبوط نحو الدعم', prob: 15, target: pw, stop: cw },
    ];
    return [
        { type: 'neutral-s', icon: '↔', name: 'Range-Bound Consolidation', nameAr: 'تداول في نطاق محدد', prob: 55, target: mp, stop: Math.round(pw * 0.995) },
        { type: 'bullish', icon: '↗', name: 'Bullish Breakout', nameAr: 'اختراق صعودي', prob: 25, target: cw, stop: zg },
        { type: 'bearish', icon: '↘', name: 'Bearish Flush to Put Wall', nameAr: 'تصحيح هبوطي نحو جدار البيع', prob: 20, target: pw, stop: Math.round(price * 1.005) },
    ];
}

function getDecision(bias, gex, price, rank) {
    if (rank < 20) return { action: '🚫 لا تداول اليوم', detail: 'IV منخفض جداً - Credit Spreads غير مجدية' };
    const dc = gex.callWall - price, dp = price - gex.putWall;
    if (bias === 'bearish' && dc < dp * 1.5) return { action: '🔴 SELL CALL Credit Spread', detail: `الجدار عند - ${Math.round(gex.zeroGamma)} الدخول عند`
    `${gex.callWall}` };
}

```

```

    if (bias === 'bullish' && dp < dc * 1.5) return { action: '🟢 SELL PUT Credit Spread', detail: `الدعم عند - ${Math.round(gex.zeroGamma)} عند الدخول عند - ${gex.putWall}` };
    return { action: '🏠 Iron Condor', detail: `و بيع بين - ${gex.putWall} و ${gex.callWall} - مغناطيس - ${gex.maxPain || Math.round(price/25)*25}` };
}

function getCalendar() {
    const now = new Date();
    return [
        { date: '2026-04-07', name: 'FOMC Meeting Minutes', nameAr: 'محضر اجتماع الفيدرالي', impact: 'HIGH', est: '-', prev: '-' },
        { date: '2026-04-08', name: 'Initial Jobless Claims', nameAr: 'طلبات الإعانة', impact: 'MEDIUM', est: '222K', prev: '219K' },
        { date: '2026-04-10', name: 'Monthly Options Expiration', nameAr: 'انتهاء خيارات شهرية', impact: 'HIGH', est: '-', prev: '-' },
    ].filter(e => new Date(e.date) >= new Date(now.toDateString()));
}

function isMarketOpen() {
    const now = new Date();
    const day = now.getUTCDay();
    const hour = now.getUTCHours();
    const min = now.getUTCMinutes();
    const totalMin = hour * 60 + min;
    if (day === 0 || day === 6) return false;
    return totalMin >= 835 && totalMin <= 1200; // 13:55-20:00 UTC = 9:30-16:00 ET
}

// =====
//   UI UPDATE
// =====

function fmt(n, d = 2) {
    if (!n && n !== 0) return '-';
    return Number(n).toLocaleString('en-US', { minimumFractionDigits: d, maximumFractionDigits: d });
}

function updateUI(data) {
    if (!data) return;
    cachedData = data;

    const now = new Date();
    document.getElementById('last-update').textContent = `آخر تحديث: ${now.toLocaleTimeString('ar-SA')}`;
}

```

```

const marketOpen = isMarketOpen();
const msEl = document.getElementById('market-status');
if (msEl) msEl.innerHTML = marketOpen
    ? '<span style="color:#00c853;font-size:11px;">🟢 السوق مفتوح</span>'
    : '<span class="market-closed">🔴 السوق مغلق - آخر بيانات</span>';

// Price
const price = data.spx?.price;
const priceEl = document.getElementById('spx-price');
if (price) {
    priceEl.textContent = fmt(price);
    if (data.spx?.isStale) priceEl.style.color = '#ff9800';
    else priceEl.style.color = '#00d4ff';
}

// GEX
document.getElementById('call-wall').textContent = fmt(data.gex?.callWall, 0) ||
'—';
document.getElementById('put-wall').textContent = fmt(data.gex?.putWall, 0) ||
'—';
document.getElementById('zero-gamma').textContent = fmt(data.gex?.zeroGamma, 0)
|| '—';
document.getElementById('max-pain').textContent = fmt(data.gex?.maxPain, 0) ||
'—';
document.getElementById('gamma-flip').textContent = fmt(data.gex?.gammaFlip, 0)
|| '—';

// Expected Move
document.getElementById('daily-upper').textContent =
fmt(data.expectedMove?.dailyUpper, 0);
document.getElementById('daily-lower').textContent =
fmt(data.expectedMove?.dailyLower, 0);
document.getElementById('weekly-upper').textContent =
fmt(data.expectedMove?.weeklyUpper, 0);
document.getElementById('weekly-lower').textContent =
fmt(data.expectedMove?.weeklyLower, 0);
if (data.expectedMove?.expDate) document.getElementById('exp-date').textContent
= `EXP: ${data.expectedMove.expDate}`;

// VIX
const vix = data.vix?.value;
if (vix) {
    document.getElementById('vix-value').textContent = fmt(vix, 2);
    const ch = data.vix.change || 0;
    const pct = data.vix.changePct || 0;
    const el = document.getElementById('vix-change');
    el.textContent = `${ch >= 0 ? '↑' : '↓'} ${fmt(Math.abs(ch), 2)}%`;
}

```



```

({fmt(Math.abs(pct), 2)}%)`;
    el.style.background = ch >= 0 ? 'rgba(255,23,68,0.15)' :
'rgba(0,200,83,0.15)';
    el.style.color = ch >= 0 ? '#ff1744' : '#00c853';
}

// IV
const rank = data.iv?.rank || 0;
const pct = data.iv?.percentile || 0;
document.getElementById('iv-rank-val').textContent = `${rank}%`;
document.getElementById('iv-rank-bar').style.width = `${Math.min(rank, 100)}%`;
document.getElementById('iv-pct-val').textContent = `${pct}%`;
document.getElementById('iv-pct-bar').style.width = `${Math.min(pct, 100)}%`;

// Strategy
const s = data.strategy || {};
document.getElementById('strategy-en').textContent = s.en || '-';
document.getElementById('strategy-ar').textContent = s.ar || '-';

// Put/Call
const pc = data.putCall?.ratio;
if (pc) {
    document.getElementById('pc-ratio').textContent = fmt(pc, 2);
    let sig, icon;
    if (pc > 1.2) { sig = '🐻 تشاؤم شديد'; icon = '😱'; }
    else if (pc > 0.9) { sig = '🐻 تشاؤم - Bearish'; icon = '📉'; }
    else if (pc < 0.6) { sig = '🐂 تفاؤل مفرط'; icon = '🚀'; }
    else if (pc < 0.75) { sig = '🐂 تفاؤل - Bullish'; icon = '📈'; }
    else { sig = '⚖️ محايد - Neutral'; icon = '📊'; }
    document.getElementById('pc-signal').textContent = sig;
    document.getElementById('pc-icon').textContent = icon;
}

// GEX Regime
const regime = data.gex?.netGex < 0 ? 'negative' : 'positive';
const rb = document.getElementById('gex-regime-box');
const rt = document.getElementById('gex-regime-text');
if (regime === 'negative') {
    rb.style.background = 'rgba(255,23,68,0.08)'; rb.style.border = '1px solid
rgba(255,23,68,0.2)';
    rt.textContent = '🔴 Negative Gamma - تقلب مرتفع'; rt.style.color = '#ff1744';
} else {
    rb.style.background = 'rgba(0,200,83,0.08)'; rb.style.border = '1px solid
rgba(0,200,83,0.2)';
    rt.textContent = '🟢 Positive Gamma - استقرار'; rt.style.color = '#00c853';
}

```

```

// Bias
const bias = data.bias || {};
const bb = document.getElementById('bias-box');
bb.className = `bias-box bias-${bias.direction || 'neutral'}`;
const bmap = { bullish: { en: 'BULLISH', ar: '📈 مساعد' }, bearish: { en: 'BEARISH', ar: '📉 هابط' }, neutral: { en: 'NEUTRAL', ar: '⚖️ محايد' } };
const bm = bmap[bias.direction] || bmap.neutral;
document.getElementById('bias-value').textContent = bm.en;
document.getElementById('bias-ar').textContent = bm.ar;
document.getElementById('bias-strength').textContent = `STRENGTH ${bias.strength || 0}%`;
document.getElementById('bias-reason').textContent = bias.reason || '';

// Trade Today
const calendar = data.calendar || [];
const highEvents = calendar.filter(e => e.impact === 'HIGH');
const tdEl = document.getElementById('trade-today');
if (highEvents.length > 0) {
    tdEl.className = 'trade-today trade-no';
    tdEl.textContent = `⚠️ تحذير - أخبار عالية التأثير | ${highEvents[0].name}`;
} else if ((data.iv?.rank || 0) < 20) {
    tdEl.className = 'trade-today trade-no';
    tdEl.textContent = `⚠️ IV منخفض جداً - لا تتداول`;
} else {
    tdEl.className = 'trade-today trade-yes';
    tdEl.textContent = `✅ Trade Today | تداول اليوم - الظروف مناسبة`;
}

// Decision
const dec = data.decision || {};
document.getElementById('decision-value').textContent = dec.action || '-';
document.getElementById('decision-detail').textContent = dec.detail || '-';

// Judas
const judas = data.judas?.active;
const jb = document.getElementById('judas-bar');
jb.className = `judas-bar ${judas ? 'active' : 'inactive'}`;
document.getElementById('judas-text').textContent = judas ? `⚠️ تحذير انحراف (جوداس)` : `🛑 لا يوجد انحراف خادع (جوداس)`;
document.getElementById('judas-text-en').textContent = judas ? 'JUDAS SWING ACTIVE - Wait for confirmation' : 'No Judas Swing detected';

// Liquidity
const liq = data.liquidity || {};
document.getElementById('pdh').textContent = fmt(liq.pdh, 2) || '-';
document.getElementById('pdl').textContent = fmt(liq.pdl, 2) || '-';
document.getElementById('weekly-high').textContent = fmt(liq.weeklyHigh, 2) ||

```

```

'-';
    document.getElementById('weekly-low').textContent = fmt(liq.weeklyLow, 2) ||
'-';

// Scenarios
const sc = document.getElementById('scenarios-container');
sc.innerHTML = (data.scenarios || []).map(s => `
    <div class="scenario-item ${s.type}">
        <div class="scenario-header">
            <span class="scenario-name">${s.icon} ${s.name} - ${s.nameAr}</span>
            <span class="scenario-prob">${s.prob}%</span>
        </div>
        <div class="scenario-bar-bg"><div class="scenario-bar-fill"
style="width:${s.prob}%"></div></div>
        <div class="scenario-details">
            <div class="scenario-target">🎯 Target: ${fmt(s.target, 0)}</div>
            <div class="scenario-stop">🛑 Stop: ${fmt(s.stop, 0)}</div>
        </div>
    </div>
`).join('');

// Calendar
const cc = document.getElementById('calendar-container');
if (!calendar.length) {
    cc.innerHTML = '<div style="color:#5a7a9a;padding:12px;text-align:center;font-size:12px;">✅ لا توجد أحداث مهمة هذا الأسبوع</div>';
} else {
    cc.innerHTML = `<table class="cal-table"><thead><tr><th>التاريخ</th>
<th>الحدث</th><th>التأثير</th><th>السابق / المتوقع</th></tr></thead><tbody>
    ${calendar.map(e => `<tr>
        <td style="color:#7a9ab0;">${e.date}</td>
        <td><div style="font-weight:600;">${e.name}</div><div
style="color:#5a7a9a;font-size:10px;">${e.nameAr}</div></td>
        <td><span class="impact-${e.impact.toLowerCase()}">${e.impact}</span></td>
        <td style="color:#7a9ab0;">${e.prev||'-'} / ${e.est||'-'}</td>
    </tr>`).join('')}
    </tbody></table>`;
}

// Key Levels
const levels = [];
if (data.gex?.callWall) levels.push({ price: data.gex.callWall, name: 'Call Wall / Major Resistance', nameAr: 'جدار الشراء - مقاومة رئيسية', color: '#00c853', s: 'strong' });
if (liq.pdh) levels.push({ price: liq.pdh, name: 'Previous Day High (PDH)', nameAr: 'أعلى أمس', color: '#ff9800', s: 'moderate' });
if (data.gex?.maxPain) levels.push({ price: data.gex.maxPain, name: 'Max Pain /

```

```

Magnet', nameAr: 'نقطة الألم الأقصى', color: '#29b6f6', s: 'strong' });
    if (data.gex?.zeroGamma) levels.push({ price: data.gex.zeroGamma, name: 'Zero-
Gamma Level', nameAr: 'مستوى الجاما المفردية', color: '#ff9800', s: 'moderate' });
    if (data.gex?.gammaFlip) levels.push({ price: data.gex.gammaFlip, name: 'Gamma
Flip Level', nameAr: 'نقطة تحول الجاما', color: '#e040fb', s: 'strong' });
    if (liq.pdl) levels.push({ price: liq.pdl, name: 'Previous Day Low (PDL)',
nameAr: 'أدنى أمس', color: '#29b6f6', s: 'moderate' });
    if (data.gex?.putWall) levels.push({ price: data.gex.putWall, name: 'Put Wall /
Major Support', nameAr: 'جدار البيع - دعم رئيسي', color: '#ff1744', s: 'strong'
});
    levels.sort((a, b) => b.price - a.price);
    document.getElementById('key-levels-container').innerHTML = levels.map(l => `
    <div class="key-level-row">
        <span class="level-price" style="color:${l.color}">${fmt(l.price, 0)}</span>
        <div class="level-name">${l.name}<br><span
style="color:#5a7a9a;">${l.nameAr}</span></div>
        <span class="level-strength ${l.s}">${l.s.toUpperCase()}</span>
    </div>
    `).join('');
}

```

```

// =====
//   MAIN LOAD
// =====

```

```

async function loadAllData() {
    // Start timer
    clearInterval(refreshTimer);
    nextRefreshTime = Date.now() + 15 * 60 * 1000;
    refreshTimer = setInterval(() => {
        const rem = Math.max(0, nextRefreshTime - Date.now());
        const m = Math.floor(rem / 60000), s = Math.floor((rem % 60000) / 1000);
        document.getElementById('next-refresh').textContent = `التحديث القادم:
${m}:${s.toString().padStart(2, '0')}`;
        if (rem <= 0) loadAllData();
    }, 1000);

    try {
        // Step 1: Get SPX price first
        const spxData = await fetchSPX();
        const price = spxData.price || 6582;

        // Step 2: Fetch everything else in parallel
        const [vixData, optionsData, liquidityData] = await Promise.allSettled([
            fetchVIX(),
            fetchOptions(price),
            fetchLiquidity(),

```

```

    });

    const vix = vixData.value || { value: 23.87, change: 0, changePct: 0, low52:
12, high52: 45, percentile: 60 };
    const options = optionsData.value;
    const liquidity = liquidityData.value || {};

    const vixVal = vix.value || 23.87;
    const rank = ivRank(vixVal, vix.low52, vix.high52);
    const pct = vix.percentile || Math.round(rank * 0.85);
    const gex = options?.gex || fallbackGEX(price);
    const maxPain = options?.maxPain || Math.round(price / 25) * 25;
    const putCall = options?.putCall || { ratio: 0.87 };

    gex.maxPain = maxPain;

    const strategy = getStrategy(rank);
    const bias = getBias(price, gex, vixVal);
    const em = getExpectedMove(price, vixVal);
    const scenarios = getScenarios(price, gex, bias.direction);
    const decision = getDecision(bias.direction, gex, price, rank);
    const calendar = getCalendar();

    const hour = new Date().getUTCHours() + 3;
    const judas = { active: (hour >= 9 && hour < 11) || (hour >= 16 && hour < 17)
};

    updateUI({
      spx: spxData,
      vix,
      gex,
      iv: { rank, percentile: pct },
      putCall,
      strategy,
      bias,
      expectedMove: em,
      scenarios,
      calendar,
      liquidity,
      decision,
      judas,
    });

  } catch (err) {
    console.error('Load error:', err);
    // Show cached data if available
    if (cachedData) updateUI(cachedData);
  }
}

```

```
    }  
}
```

```
// Start  
loadAllData();  
</script>  
</body>  
</html>
```